

LBP (Local Binary Pattern) works by looking at each pixel and comparing it to its 8 neighbors. If a neighbor is $\geq$ the center pixel, write 1; otherwise 0. Reading these 8 bits clockwise gives you a binary number (0–255) — that's the LBP code for that pixel. Do this for every pixel, and you get a texture "fingerprint."

```
python

import numpy as np
from skimage.feature import local_binary_pattern
from skimage import data
import matplotlib.pyplot as plt

# Load a sample image (grayscale)
image = data.camera()          # 512×512 grayscale photo

# Compute LBP
radius = 1                    # how far neighbors are from center
n_points = 8                  # how many neighbor points (usually 8)
lbp = local_binary_pattern(image, n_points, radius, method='uniform')

# Plot
fig, axes = plt.subplots(1, 3, figsize=(12, 4))

axes[0].imshow(image, cmap='gray')
axes[0].set_title("Original image")

axes[1].imshow(lbp, cmap='gray')
axes[1].set_title("LBP output")

axes[2].hist(lbp.ravel(), bins=256, color='steelblue')
axes[2].set_title("LBP histogram (texture descriptor)")

plt.tight_layout()
plt.show()
```